Correlation-consistent Gaussian basis sets for copper solids from material-constrained atomic optimization

Jincheng Yu,[1] Xiaoyu Zhang,[1,2] Min-Ye Zhang,[3,4] Yu Cao,[5,3] Qiming Sun,[6] and Hong-Zhou Ye[1,7,a]

[1)] *Department of Chemistry and Biochemistry, University of Maryland, College Park, Maryland, 20742, United States*

[2)] *College of Chemistry and Molecular Engineering, Peking University, Beijing, 100871, China*

[3)] *Institute of Physics, Chinese Academy of Sciences, Beijing, 100190, China*

[4)] *The NOMAD Laboratory at the Fritz Haber Institute of the Max-Planck-Gesellschaft, Berlin, 14195, Germany*

[5)] *School of Physics, Peking University, Beijing, 100871, China*

[6)] *Quantum Engine LLC, Lacey, Washington 98516, United States*

[7)] *Institute for Physical Science and Technology, University of Maryland, College Park, Maryland, 20742, United States*

(Dated: 13 July 2026)

Correlation-consistent Gaussian basis sets enable systematic complete-basis-set (CBS) extrapolation in molecular quantum chemistry, but diffuse atomically optimized primitives can cause severe linear dependence in periodic solids, particularly metals. We introduce material-constrained atomic optimization (MCAO), which preserves the atomic and correlation-consistent character of Gaussian basis sets while penalizing large overlap-matrix condition numbers in representative solids. As a proof of concept, we generate Dunning-style MCAO-cc-pVXZ sets (X = D, T, Q) for Cu with nonrelativistic and scalar-relativistic all-electron, effective core potential (ECP), and pseudopotential treatments. The resulting MCAO-cc-pVXZ basis sets remain numerically stable for Cu solids and surfaces while reproducing molecular Cu dimer energetics and plane-wave reference properties of bulk Cu. CBS-extrapolated random-phase approximation calculations further enable a controlled assessment of pseudopotential, relativistic, and basis-set errors in bulk Cu and CO adsorption on Cu(111), providing scalar-relativistic all-electron Gaussian-basis benchmarks for the CO adsorption puzzle.

Correlated-wavefunction methods are becoming increasingly useful for benchmark calculations of solids and surfaces,[1–6] where chemically relevant energy differences can depend sensitively on exchange,[7,8] correlation,[9–12] finite-size effects,[13–15] and basis-set convergence.[16–19] These requirements are especially demanding for transition-metal systems, where localized d electrons, metallic screening, and adsorption-induced charge rearrangements must be described on the same footing.[20] The CO adsorption puzzle on transition-metal surfaces[21–23] is a representative example: small site-preference energies distinguish the experimentally observed top site from the hollow site favored by semilocal density functional theory (DFT).[21,22,24] For such problems, reliable complete-basis-set (CBS) extrapolation is essential for separating electronic-structure errors from basis-set artifacts.

Gaussian basis sets are attractive for this purpose because molecular quantum chemistry has developed systematic correlation-consistent families that enable reliable CBS extrapolation in both mean-field and correlated-wavefunction calculations.[25–27] However, standard Gaussian basis sets optimized for isolated atoms often contain diffuse primitives that become nearly linearly dependent in periodic solids.[19,28–30] Practical remedies, including exponent truncation[31,32] and system-specific optimization,[17,18,30,33–35] can improve numerical stability but may compromise transferability or obscure systematic convergence toward the CBS limit.[17,36] A solid-state Gaussian basis set should therefore retain the accuracy and correlation-consistent structure of atomic optimization while controlling linear dependence in representative materials.

In this work, we address this need by introducing material-constrained atomic optimization (MCAO), which augments standard atomic basis-set optimization with a material-dependent stability constraint. MCAO enables continuous tuning of basis-set linear dependence while largely preserving the atomic basis-set structure, including the correlation consistency needed for correlated-wavefunction calculations.[26,37] As a proof of concept, we apply MCAO to generate Dunning-style correlation-consistent Cu basis sets, denoted MCAO-cc-pVXZ (X = D, T, Q), for a range of nuclear-potential and relativistic treatments. The resulting basis sets remain numerically stable for Cu-containing solids while reproducing molecular Cu dimer energetics and plane-wave reference properties of bulk Cu. CBS-extrapolated random-phase approximation[38–41] (RPA) calculations further enable a controlled assessment of pseudopotential, relativistic, and basis-set errors in bulk Cu and CO adsorption on Cu(111), providing scalar-relativistic all-electron Gaussian-basis benchmarks for the CO adsorption puzzle.

Most widely used Gaussian basis sets are generated by atomic optimization,

$$\boldsymbol{\alpha} \leftarrow \min_{\boldsymbol{\alpha}} E_{\mathrm{atom}}(\boldsymbol{\alpha}) \tag{1}$$

where $\boldsymbol{\alpha} = (\boldsymbol{\alpha}^{l=0}, \boldsymbol{\alpha}^{l=1}, \cdots)$ denotes the primitive Gaussian exponents grouped by angular momentum l. The atomic ob-

[a)] Electronic mail: hzye@umd.edu

jective $E_{\rm atom}$ may be a mean-field or a correlation energy. For example, the correlation-consistent cc-pVXZ basis sets introduced by Dunning[37] combine a valence set optimized at the HF level with a polarization set optimized at a correlated level, both following eq. (1). Other prominent examples include the def2 series of Weigend and Ahlrichs,[42,43] the polarization-consistent (pc) series of Jensen,[44] and extensions or augmentations designed for extended electronic states,[45] core and semicore correlation,[46] and relativistic corrections.[47,48] Together, these developments provide a rich library of Gaussian basis sets for highly accurate molecular quantum chemistry.[27]

Material-constrained atomic optimization (MCAO) is designed to retain the accuracy and transferability of atomic optimization while addressing one of its main shortcomings for condensed-phase calculations, namely that unconstrained atomic optimization can generate diffuse primitives that lead to severe linear dependence in bulk solids and surfaces. This problem is especially acute for metallic and metal-containing systems, where the lower effective nuclear charge of metal atoms often produces very diffuse primitives, for example with $\alpha \ll 0.1$ a.u..[19,49] Figure 1 illustrates this issue for several atomically optimized basis sets of Cu, including the cc-pVXZ series,[50] the def2-XZVP series,[43] and the ccECP-cc-pVXZ series[51] developed for the correlation-consistent effective core potential[51,52] (ccECP). We characterize the linear dependence of each basis by the condition number κ of the overlap matrix evaluated for face-centered cubic (*fcc*) Cu. Previous work has shown that $\kappa \gtrsim 10^9$ often causes numerical instabilities, including difficult self-consistent field (SCF) convergence and nonsmooth potential energy surfaces.[19] By this criterion, all atomically optimized basis sets considered in fig. 1 are likely to be unstable for periodic Cu calculations. Consistent with this observation, previous Gaussian-basis calculations on Cu surfaces have used compact or truncated basis sets, including the removal of small-exponent primitives, to avoid SCF instabilities.[31,53]

Linear dependence in Gaussian basis sets has been recognized since early developments of Gaussian-based HF methods for periodic solids.[31,32] Prior regularization strategies include discarding primitives with small exponents,[32,49] performing solid-specific[17,18,30,34] or molecule-specific[33,35,54] optimizations, and limiting the size of the valence set.[19] In most cases, however, the atomic optimization protocol and the associated basis-set structure, such as correlation consistency, are either heavily modified or abandoned. As a result, the resulting solid-friendly basis sets can have limited transferability[17,30,34,36] or may not be suitable for correlated calculations.[19] One exception is the recently reported GTH-cc-pVXZ basis sets optimized for the Goedecker-Teter-Hutter (GTH) pseudopotentials for elements from H to Ar.[19] These basis sets are atomically optimized and largely preserve the correlation-consistency structure, while controlling linear dependence by judiciously limiting the valence-set size. Extensive benchmarks have validated the accuracy of the GTH-cc-pVXZ sets for diverse materials and levels of theory,[12,15,55–60] highlighting the value of retaining key features of atomic optimization. The discrete nature of the valence-set selection protocol, however, raises the question of whether accuracy and

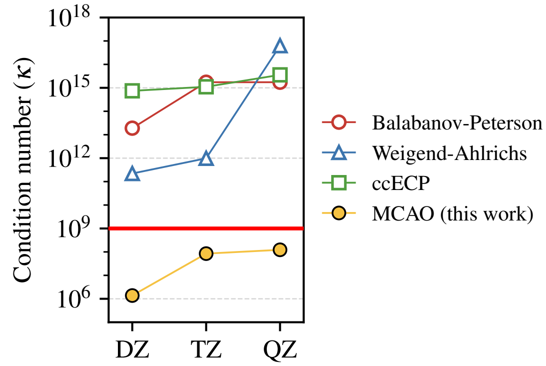

FIG. 1. Basis-set condition number κ for *fcc* Cu at the equilibrium lattice constant (3.615 Å) using different Gaussian basis sets: the cc-pVDZ/TZ/QZ sets of Balabanov and Peterson,[50] the def2-SVP/TZVP/QZVP sets of Weigend and Ahlrichs,[43] the ccECP-cc-pVDZ/TZ/QZ sets developed for the ccECP,[51] and the MCAO-cc-pVDZ/TZ/QZ sets developed in this work. The condition number is evaluated from the union of k-points in $n \times n \times n$ Γ-centered meshes with $n = 1, 2, \cdots, 5$.

numerical stability can always be balanced through this route.

MCAO takes a different route. Its design principle is to preserve the standard atomic optimization as much as possible while adding the minimal regularization needed for numerical stability in solids. This leads to the following MCAO loss function for optimizing Gaussian exponents:

$$\mathcal{L}(\boldsymbol{\alpha};\{M\},\kappa_0,\varepsilon_0) = E_{\rm atom}(\boldsymbol{\alpha}) + \frac{\varepsilon_0}{\kappa_0} \times \max_{M\in\{M\}} \kappa(\boldsymbol{\alpha};M). \quad (2)$$

This objective regularizes the standard atomic energy minimization by penalizing large condition numbers on a set of reference solids $\{M\}$ containing the element of interest. The κ penalty is applied on a linear scale and is controlled by two user-selected parameters: κ_0, which is related to the target condition number, and ε_0, which sets the energy scale of the penalty. Strictly speaking, only the ratio ε_0/κ_0 is required to define eq. (2); we retain both parameters to make their distinct roles explicit. In practice, we find it effective to fix ε_0 to the characteristic energy scale of the chosen $E_{\rm atom}$ and tune the penalty strength through κ_0. Unlike exponent truncation, MCAO determines the appropriate compactness through optimization rather than by imposing a rigid exponent cutoff.[31,49,53] Because the dominant term remains an atomic energy, the resulting basis sets are expected to be less reference-dependent than fully solid-optimized bases.[17,30,34] MCAO also enables continuous tuning of the condition number on the chosen reference materials, avoiding the coarser control associated with discrete valence-set selection.[19]

As a proof of concept, we use MCAO to generate correlation-consistent Gaussian basis sets for Cu up to quadruple-zeta (QZ) quality, hereafter denoted MCAO-cc-pVXZ. Following established recipes for molecular Cu basis sets,[43] the atomic optimization part of MCAO minimizes the HF energy for the valence set and the MP2 correlation energy for the polarization set. Both atomic energies are evaluated for the [Ar]$3d^{10}4s^1$ configuration of Cu, and the MP2 calculation correlates the 11 valence electrons in the $3d$ and $4s$ shells. Because the valence $4p$ shell is unoccupied at the HF level, we follow ref 19 and optimize its exponents in the same manner as the polarization functions.

The MCAO κ penalty is evaluated for *fcc* Cu at the experimental lattice constant (3.615 Å). To account for the different energy scales of HF and MP2 correlation energies, we use $\varepsilon_0 = 1$, 0.1, and 0.01 milli-Hartree for the valence set, the $4p$ shell, and the polarization set, respectively. The smaller ε_0 for the polarization set helps preserve the correlation-consistency structure, which is essential for the correlated calculations discussed below. For each chosen ε_0, we initialize MCAO with κ_0 slightly above the condition number of the input basis and then progressively reduce κ_0 to a target value of 10^9. In standard atomic optimization, the valence and polarization sets are optimized independently and each optimization is performed once. In MCAO, the κ penalty couples all exponent subsets because it depends on the full basis, so the subset optimizations are performed iteratively until convergence. In practice, we found that three macrocycles are sufficient to converge both the loss function and the basis exponents to reasonable precision.

After exponent optimization, the valence primitives are contracted using spherically averaged HF atomic natural orbitals (ANOs).[61] We then partially decontract the lowest one, two, and three exponents to form the DZ, TZ, and QZ valence sets, respectively. We note that HF ANOs yield fewer contracted valence orbitals than the atomic cc-pVXZ sets at the same zeta level because the latter use ANOs derived from configuration interaction.[50] For Cu, the use of HF ANOs is justified by the closed $3d$ shell in the atomic ground state, which has minimal multireference character.[43] For polarization functions, we use a hierarchical $1f$, $2f1g$, and $3f2g1h$ structure for DZ, TZ, and QZ, respectively. The TZ and QZ polarization spaces match those of the atomic cc-pVTZ and cc-pVQZ sets,[50] whereas the atomic cc-pVDZ set uses a $2f \to 1f$ contraction in which the same two f primitives from the TZ set are contracted into one f function. Benchmarks on molecules and solids show no noticeable difference between these two DZ polarization choices. We therefore adopt the simpler $1f$ structure, which also matches the widely used molecular def2-SVP set.[43] Combining the valence and polarization spaces at each XZ level gives the final MCAO-cc-pVXZ basis sets.

We apply this protocol to generate MCAO-cc-pVXZ sets for several nuclear-potential and relativistic treatments: the all-electron (AE) potential, hard and soft ccECPs,[51,62] and small-core (sc) and large-core (lc) GTH-PBE pseudopotentials.[63] For the AE potential, we generate basis sets for both nonrelativistic (NR) calculations and calculations with the one-electron spin-free exact-two-component[64–66] (SFX2C-1e) scalar relativistic correction. In MCAO, changing the nuclear potential or relativistic treatment primarily affects the atomic energy in eq. (2) and the HF ANOs used for contraction, while the solid-state κ penalty remains unchanged. To account for differences in core size and potential hardness, we adjust the size of the valence primitive set while matching the two or three most diffuse exponents in each angular-momentum channel across nuclear-potential and relativistic treatments. A similar exponent matching is achieved automatically for the polarization set by using the same hierarchical structure described above. These choices maintain a consistent valence-electron basis-set quality across the different nuclear-potential and relativistic treatments.

TABLE I. Nonrelativistic RPA@PBE binding energy of a Cu dimer at a separation of 2.15 Å calculated with different Gaussian basis sets. The def2-TZVP* basis is obtained from def2-TZVP by removing primitives with exponents below 0.05 a.u., following ref 53.

Basis set	E_{bind} (eV/atom)
cc-pVDZ	−0.70
cc-pVTZ	−0.72
cc-pVQZ	−0.75
CBS(T,Q)	−0.75
MCAO-cc-pVDZ	−0.69
MCAO-cc-pVTZ	−0.71
MCAO-cc-pVQZ	−0.73
CBS(T,Q)	−0.74
def2-TZVP*	−0.86
pob-TZVP	−2.37

The optimized exponents of the MCAO-cc-pVQZ sets for all six nuclear-potential and relativistic treatments are shown in fig. 2, illustrating the exponent matching in both the valence and polarization spaces. Compared with the standard atomic cc-pVQZ set, the MCAO-cc-pVQZ sets have a more compact valence space, leading to substantially improved numerical stability in solids. This improvement is reflected in the reduced κ values for the AE/NR MCAO sets in fig. 1 and for the other nuclear-potential and relativistic treatments in Table S2. Importantly, the polarization exponents of the MCAO sets remain close to those of the atomic cc-pVQZ set, indicating that the correlation-consistency structure is largely preserved.

Having established the improved numerical stability of the MCAO-cc-pVXZ sets, we next examine whether the regularization compromises accuracy. For a molecular test, table I reports nonrelativistic RPA binding energies of the Cu dimer. Throughout this work, all RPA calculations employ Kohn–Sham orbitals obtained with the Perdew–Burke–Ernzerhof (PBE) functional;[67] we denote this approach RPA@PBE. The MCAO-cc-pVXZ results agree with the standard cc-pVXZ values within 0.02 eV/atom (*ca.* 0.5 kcal/mol/atom) at each zeta level and in the extrapolated complete-basis-set (CBS) limit. This agreement is notable because the MCAO bases are simultaneously constrained to remain numerically stable for solids. By contrast, solid-friendly basis sets generated by exponent truncation, including pob-TZVP[49] and the modified def2-TZVP set of ref 53, overestimate the binding energy relative to standard cc-pVTZ, suggesting a less balanced description of the Cu atom and dimer.

The same balance between stability and accuracy is observed for periodic Cu. Figure 3 compares PBE lattice constants, bulk moduli, and band structures of *fcc* Cu calculated with the GTH-PBE-sc pseudopotential. For this Hamiltonian,

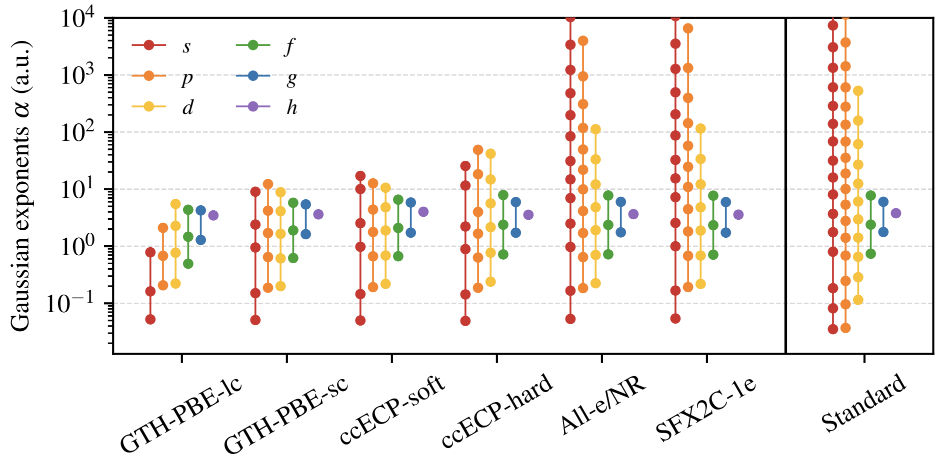

FIG. 2. Primitive Gaussian exponents of the Cu MCAO-cc-pVQZ basis sets optimized for different nuclear-potential and relativistic treatments, compared with the standard atomic cc-pVQZ basis set.[50] Colors denote angular-momentum channels. The valence space contains the s, p, and d channels, whereas the polarization space contains the f, g, and h channels. The MCAO procedure mainly compacts the most diffuse valence exponents while keeping the polarization exponents close to the atomic correlation-consistent reference.

a sufficiently converged PW calculation provides an unambiguous CBS reference. The MCAO-cc-pVXZ lattice constant (a_0) and bulk modulus (B_0) converge rapidly with zeta quality, reaching the PW reference within 0.01 Å and 1 GPa, respectively. The high accuracy in ground-state PBE calculations is consistent with an accurate description of the valence bands by the MCAO-cc-pVXZ sets. In addition, the low-energy conduction bands, which are central to response and correlated-wavefunction calculations, are also reproduced well. These results suggest that accurate structural properties and energy spectra can emerge naturally from atomically optimized Gaussian basis sets once their linear dependence is controlled. Explicit solid-specific energy optimization or band-structure fitting, as proposed in previous work,[17,18,30,34] may therefore not be necessary for designing accurate Gaussian basis sets for solids.

Reliable CBS extrapolation for correlated-wavefunction calculations in solids makes it possible to isolate errors from other approximations, including pseudopotentials, ECPs, and the neglect of relativistic effects. We use SFX2C-1e all-electron calculations, enabled by recent extensions of SFX2C-1e to periodic solids,[72] as the theoretical reference. Figure 4 compares CBS-extrapolated RPA@PBE a_0 and B_0 values for bulk Cu obtained with different nuclear-potential and relativistic treatments. The GTH-PBE-sc pseudopotential and both ccECPs are in reasonable agreement with the SFX2C-1e all-electron reference. In contrast, the GTH-PBE-lc pseudopotential substantially overestimates the Cu–Cu interaction strength, underestimating a_0 by approximately 0.1 Å and overestimating B_0 by approximately 70 GPa. This failure cannot be attributed simply to the use of a large core, because the same number of electrons are correlated in all calculations. The result instead highlights the challenge of parameterizing pseudopotentials for correlated-wavefunction calculations.[52]

Figure 4 also demonstrates why reaching the CBS limit is essential for this comparison. For the GTH-PBE-sc pseudopotential, we also report our RPA@PBE result obtained with the TZV2P-MOLOPT basis set developed by Li and co-workers[35] (optimized for the ωB97X-V functional[73]) using the MOLOPT approach.[33] Although this basis has been shown to perform well in DFT calculations of Cu-containing solids,[35] it shows a sizable basis-set incompleteness error in RPA, likely because it lacks a correlation-consistency structure. Importantly, this error should not be attributed to the GTH-PBE-sc pseudopotential itself, which is accurate at the CBS limit as demonstrated by our MCAO basis sets. The def2-SVP all-electron result provides a similar cautionary example: it appears reasonably accurate relative to the SFX2C-1e reference, but comparison with our CBS-limit nonrelativistic calculation reveals a fortuitous cancellation between basis-set incompleteness and the neglect of scalar relativistic effects.

Our converged Gaussian-basis SFX2C-1e results, $a_0 \approx$ 3.59 Å and $B_0 \approx 147$ GPa, are also consistent with independent theoretical benchmarks and experimental references. Figure 4 includes several literature results obtained with PW bases and the projector augmented wave (PAW) approximation,[74,75] including a_0 and B_0 from Harl and co-workers[68] and Nepal and co-workers,[70] as well as a_0 from Schimka and co-workers.[69] The PW/PAW framework is widely used for solid-state RPA calculations and has been adopted in recent large-scale benchmarks of RPA-based double-hybrid functionals.[76,77] Overall, our best estimate agrees with the spread of published PW/PAW values. It is also in good agreement with zero-point-energy-corrected experimental references compiled by Harl and co-workers[68] and Schimka and co-workers,[71] supporting the accuracy of RPA@PBE for structural properties of transition-metal solids.

As a final application, we revisit the CO adsorption puzzle on Cu(111). The CO puzzle refers to the failure of semilocal Kohn–Sham DFT to predict the experimentally pre-

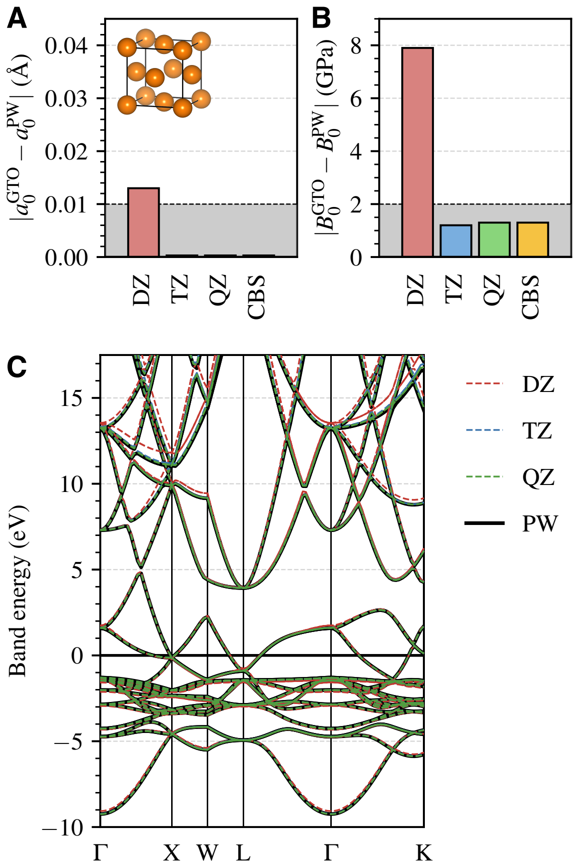

FIG. 3. PBE lattice constant, bulk modulus, and band structure of *fcc* Cu calculated with the GTH-PBE-sc pseudopotential and the associated MCAO-cc-pVXZ basis sets. Panels A and B show the absolute errors of the lattice constant and bulk modulus relative to plane-wave (PW) references obtained with the same pseudopotential. The CBS limit is obtained from a (TZ,QZ)-extrapolation. The gray shaded area indicates an error within 0.01 Å and 2 GPa. Panel C compares the MCAO-cc-pVXZ and PW PBE band structures, with band energies shifted by the Fermi energy determined in the corresponding basis.

ferred adsorption site on several transition-metal surfaces.[21,22] For Cu(111), shown in fig. 5A, semilocal DFT favors the threefold *fcc hollow* site,[21,22,24] largely because of delocalization[7,31] or density-driven[23] errors. Low-temperature experiments based on Fourier-transform reflection-absorption infrared spectroscopy[80] (FT-RAIRS) and angle-resolved photoemission extended fine structure[81] (ARPEFS), however, observe only signatures of adsorption at the onefold *top* site. Early studies by Ren and co-workers[78] and Schimka and co-workers[79] showed that RPA based on a semilocal DFT reference recovers the correct site preference. More recent beyond-RPA calculations, including quantum Monte Carlo,[14,82] coupled cluster,[83] and multireference configuration interaction,[84] have confirmed the RPA site preference and further refined the adsorption energies relative to experiment.

Most previous periodic calculations used PW bases with pseudopotentials,[24,79,82,83] whereas atom-centered local bases have mainly been explored through cluster embedding approximations.[78,84] Cao and co-workers recently reported fully periodic Gaussian-basis RPA@PBE calculations of CO@Cu(111) adsorption energies.[53] Although those calculations used relatively small modified def2-SVP and def2-TZVP basis sets and neglected relativistic effects, the reported RPA@PBE adsorption energies were in reasonable agreement with prior PW results. Here, we revisit this problem using MCAO-cc-pVXZ basis sets to remove basis-set incompleteness in a controlled way. This allows us to examine, on a common CBS-limit footing, the effects of pseudopotentials, ECPs, and scalar relativistic corrections.

Figure 5B reports CBS-extrapolated RPA@PBE adsorption energies for CO on the top and hollow sites using different nuclear-potential and relativistic treatments. All reported adsorption energies include a counterpoise correction for basis-set superposition error.[85] The procedure follows established protocols[15,55,86,87] and is detailed in the Supporting Information. Relative to the all-electron SFX2C-1e reference, the GTH-PBE-sc pseudopotential and the soft ccECP give adsorption energies within 0.05 eV for both sites. The hard ccECP underestimates both adsorption energies by *ca.* 0.05 eV, while nonrelativistic all-electron calculations underestimate them by *ca.* 0.1 eV. These errors are largely systematic across the two adsorption sites and therefore cancel in the site-preference energy,

$$\Delta E_{\mathrm{ads}} = E_{\mathrm{ads}}^{\mathrm{hollow}} - E_{\mathrm{ads}}^{\mathrm{top}} \tag{3}$$

shown in fig. 5C. Except for GTH-PBE-lc, all pseudopotential, ECP, and nonrelativistic all-electron treatments reproduce the SFX2C-1e ΔE_{ads} within 0.02 eV (*ca.* 0.5 kcal/mol). The GTH-PBE-lc pseudopotential is the clear outlier: it overbinds CO at both sites by more than 0.4 eV and predicts an adsorption-energy gap close to zero. These adsorption-energy trends mirror the bulk structural trends in fig. 4, suggesting that the errors associated with different nuclear potentials and relativistic treatments are systematic and transferable between bulk Cu and Cu surfaces.

Figure 5(B,C) also compares our results with previous RPA@PBE calculations by Ren and co-workers using an all-electron numerical atomic orbital (NAO) basis[78] and by Schimka and co-workers using PW/PAW.[79] Our SFX2C-1e adsorption energies are in reasonable agreement with the NAO results of Ren and co-workers, while the SFX2C-1e site-preference energy agrees more closely with the PW/PAW result. Because all three studies made substantial efforts to approach the CBS limit, the remaining differences likely arise from other sources, including relativistic treatments, slab models, and pseudopotential approximations. A definitive disentanglement of these errors will require coordinated cross-code comparisons, which are left for future work. The MCAO-cc-pVXZ sets developed here provide a controlled Gaussian-basis route for such comparisons because the basis-set error can be systematically reduced while maintaining numerical stability.

We also comment on the fully periodic Gaussian-basis RPA@PBE calculations of Cao and co-workers, who reported adsorption energies of −0.31 and −0.19 eV for the top and hollow sites, respectively, corresponding to a site-preference energy of 0.11 eV.[53] These values are in reason-

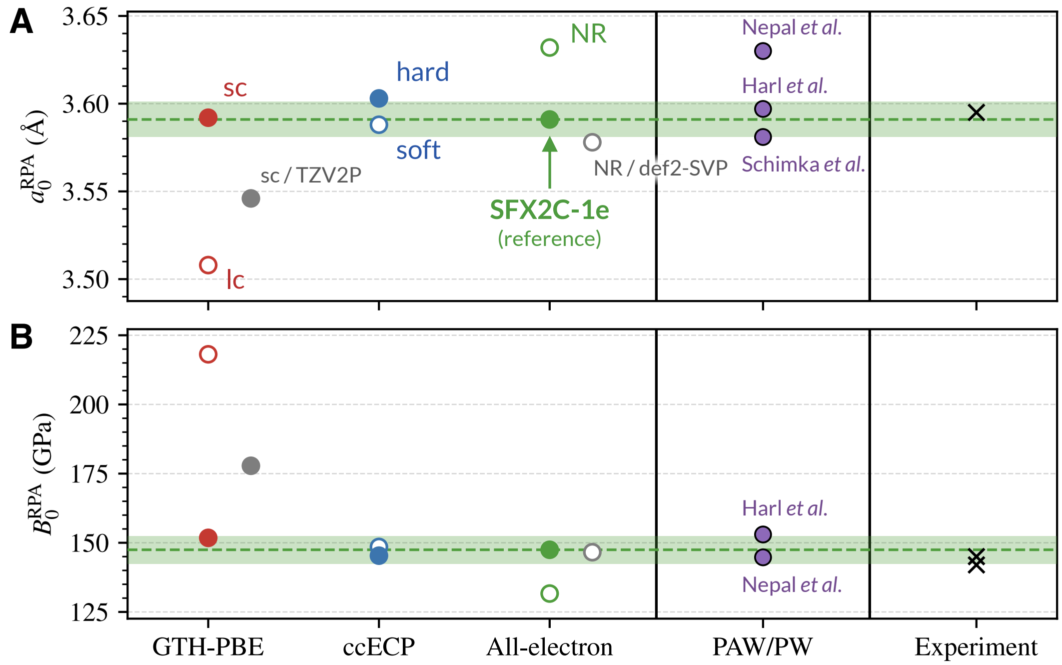

FIG. 4. RPA@PBE lattice constant (top) and bulk modulus (bottom) of *fcc* Cu extrapolated to the CBS limit using MCAO-cc-pV(T,Q)Z basis sets for different nuclear-potential and relativistic treatments. The SFX2C-1e all-electron result is used as the theoretical reference; the green shaded region indicates deviations within ±0.01 Å for a_0 and ±5 GPa for B_0. For the GTH-PBE-sc pseudopotential, the filled gray circle shows our result obtained with the TZV2P-MOLOPT basis set developed by Li and co-workers.[35] For the all-electron potential, the hollow gray circle shows the nonrelativistic def2-SVP result. Literature PW/PAW results from Harl *et al.*,[68] Schimka *et al.*,[69] and Nepal *et al.*[70] and zero-point-energy-corrected experimental references[68,71] are included for comparison.

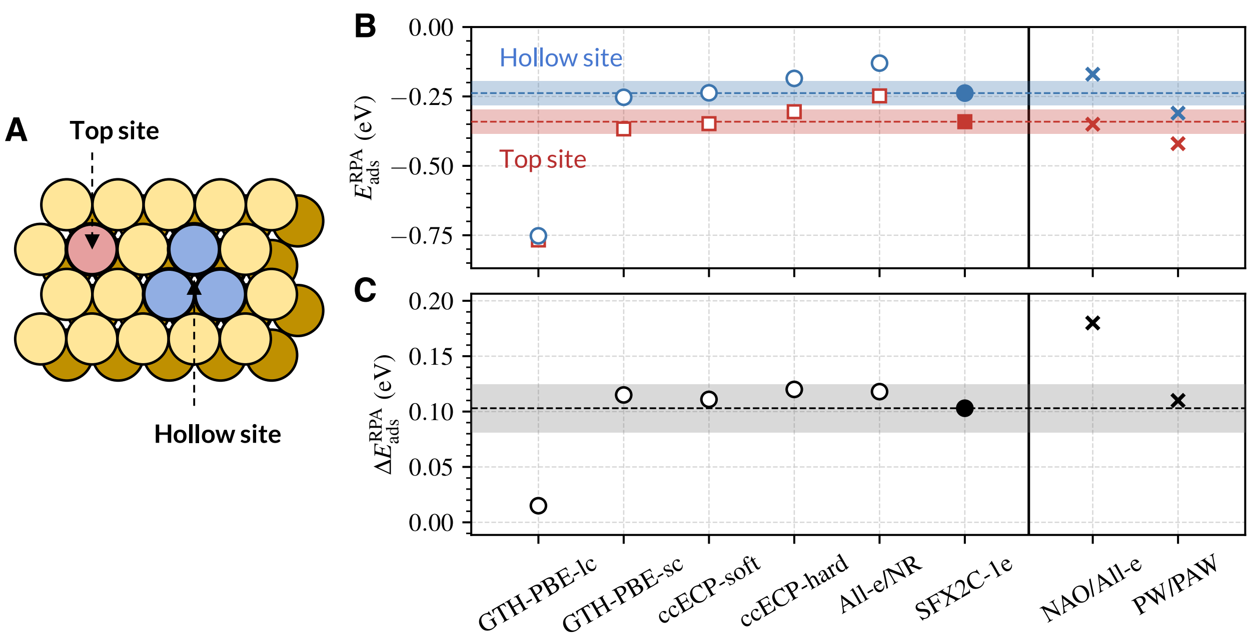

FIG. 5. (A) Cu(111) surface model and the top and hollow sites for CO adsorption. (B) CBS-extrapolated RPA@PBE adsorption energies of CO on top and hollow sites of Cu(111), obtained using MCAO-cc-pV(T,Q)Z basis sets with counterpoise correction for different nuclear-potential and relativistic treatments. The shaded region indicates deviations smaller than ±1 kcal/mol (*ca.* 0.04 eV) from the SFX2C-1e reference. Literature results from Ren and co-workers using a numerical atomic orbital (NAO) basis[78] and from Schimka and co-workers using PW/PAW[79] are included for comparison. (C) RPA@PBE adsorption-energy difference defined in eq. (3). The shaded region indicates deviations smaller than ±0.5 kcal/mol (*ca.* 0.02 eV) from the SFX2C-1e reference.

able agreement with our SFX2C-1e reference and with the

PW/PAW results of Schimka and co-workers.[79] However, we found that reproducing this comparison is sensitive to both basis-set extrapolation and finite-size extrapolation to the thermodynamic limit. As discussed in the Supporting Information, our calculations using the same modified def2-SVP and def2-TZVP basis sets give substantially less favorable CBS(SVP,TZVP)-extrapolated adsorption energies, while an analogous CBS(DZ,TZ) extrapolation with nonrelativistic all-electron MCAO-cc-pVXZ basis sets gives similar values. This consistency suggests that the remaining discrepancy is primarily associated with the absence of QZ-quality information in a DZ/TZ-only extrapolation, together with differences in the treatment of finite-size effects, rather than with the use of Gaussian basis sets per se.

To conclude, we have introduced material-constrained atomic optimization as an effective strategy for generating Gaussian basis sets that retain the accuracy of atomic optimization while remaining numerically stable for periodic solids. We applied MCAO to construct correlation-consistent Cu basis sets for several pseudopotentials, ECPs, and all-electron Hamiltonians, including nonrelativistic and SFX2C-1e scalar relativistic treatments. The resulting MCAO-cc-pVXZ basis sets reproduce Cu dimer binding energies from standard cc-pVXZ sets and accurately recover PW reference structural properties and band structures for bulk Cu. Their reliable CBS extrapolation enables a controlled assessment of pseudopotential, ECP, and relativistic errors in RPA@PBE calculations of bulk Cu. We further applied these basis sets to CO adsorption on Cu(111), obtaining scalar-relativistic all-electron RPA@PBE benchmarks for the top and hollow adsorption sites.

The present MCAO formulation also has several limitations that point to natural directions for future development. First, basis-set linear dependence is often associated with multiple ill-conditioned modes of the overlap matrix, whereas the condition-number penalty used here depends only on the most extreme eigenvalues. A more flexible regularization could penalize all near-singular modes, for example through a smooth threshold applied to the small-eigenvalue spectrum. Second, the current MCAO scheme controls linear dependence only through the primitive exponents. Future work could extend the optimization to the contraction coefficients, an avenue already exploited in the MOLOPT basis sets to improve numerical stability.[33,35] Third, extending MCAO beyond Cu to other transition-metal and main-group solids may require element-specific refinements, because both the atomic electronic structure and the solid-state bonding environment can change the balance between accuracy and stability. Nevertheless, the central idea of combining atomic accuracy with material-dependent stability constraints should provide a useful foundation for future solid-state Gaussian basis-set optimization.

SUPPORTING INFORMATION

The Supporting Information contains (i) geometry and basis-set data files; (ii) computational details; (iii) additional details of MCAO basis-set generation and construction of the corresponding auxiliary basis sets for density fitting; (iv) convergence of the bulk Cu and CO/Cu(111) calculations to the CBS and thermodynamic limits; and (v) a comparison with the modified def2 basis sets for CO/Cu(111) adsorption.

CONFLICT OF INTEREST

The authors declare no competing financial interest.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ACKNOWLEDGMENTS

This work was supported by the National Science Foundation under Grant No. CHE-2543461. We acknowledge computing resources provided by the Division of Information Technology at the University of Maryland, College Park.

[1] S. Svelle, C. Tuma, X. Rozanska, T. Kerber, and J. Sauer, "Quantum chemical modeling of zeolite-catalyzed methylation reactions: Toward chemical accuracy for barriers," J. Am. Chem. Soc. **131**, 816–825 (2009).
[2] C. Müller and B. Paulus, "Wavefunction-based electron correlation methods for solids," Phys. Chem. Chem. Phys. **14**, 7605–7614 (2012).
[3] G. H. Booth, A. Grüneis, G. Kresse, and A. Alavi, "Towards an exact description of electronic wavefunctions in real solids," Nature **493**, 365–370 (2013).
[4] F. Libisch, C. Huang, and E. A. Carter, "Embedded correlated wavefunction schemes: Theory and applications," Acc. Chem. Res. **47**, 2768–2775 (2014).
[5] J. McClain, Q. Sun, G. K.-L. Chan, and T. C. Berkelbach, "Gaussian-based coupled-cluster theory for the ground-state and band structure of solids," J. Chem. Theory Comput. **13**, 1209–1218 (2017).
[6] I. Y. Zhang and A. Grüneis, "Coupled cluster theory in materials science," Front. Mater. **6**, 123 (2019).
[7] A. Stroppa, K. Termentzidis, J. Paier, G. Kresse, and J. Hafner, "Co adsorption on metal surfaces: A hybrid functional study with plane-wave basis set," Phys. Rev. B **76**, 195440 (2007).
[8] H.-R. Liu, H. Xiang, and X. G. Gong, "First principles study of adsorption of o2 on al surface with hybrid functionals," J. Chem. Phys. **135**, 214702 (2011).
[9] J. Yang, W. Hu, D. Usvyat, D. Matthews, M. Schütz, and G. K.-L. Chan, "Ab initio determination of the crystalline benzene lattice energy to sub-kilojoule/mole accuracy," Science **345**, 640–643 (2014).
[10] N. Masios, A. Irmler, T. Schäfer, and A. Grüneis, "Averting the infrared catastrophe in the gold standard of quantum chemistry," Phys. Rev. Lett. **131**, 186401 (2023).
[11] V. A. Neufeld and T. C. Berkelbach, "Highly accurate electronic structure of metallic solids from coupled-cluster theory with nonperturbative triple excitations," Phys. Rev. Lett. **131**, 186402 (2023).
[12] H.-Z. Ye and T. C. Berkelbach, "Periodic local coupled-cluster theory for insulators and metals," J. Chem. Theory Comput. **20**, 8948–8959 (2024).
[13] T. Gruber, K. Liao, T. Tsatsoulis, F. Hummel, and A. Grüneis, "Applying the coupled-cluster ansatz to solids and surfaces in the thermodynamic limit," Phys. Rev. X **8**, 021043 (2018).
[14] G. R. Iyer and B. M. Rubenstein, "Finite-size error cancellation in diffusion monte carlo calculations of surface chemistry," J. Phys. Chem. A **126**, 4636–4646 (2022).
[15] H.-Z. Ye and T. C. Berkelbach, "Adsorption and vibrational spectroscopy of co on the surface of mgo from periodic local coupled-cluster theory," Faraday Discuss. **254**, 628–640 (2024).

[16] T. Olsen and K. S. Thygesen, "Random phase approximation applied to solids, molecules, and graphene-metal interfaces: From van der waals to covalent bonding," Phys. Rev. B **87**, 075111 (2013).
[17] M. A. Morales and F. D. Malone, "Accelerating the convergence of auxiliary-field quantum monte carlo in solids with optimized gaussian basis sets," J. Chem. Phys. **153**, 194111 (2020).
[18] V. A. Neufeld, H.-Z. Ye, and T. C. Berkelbach, "Ground-state properties of metallic solids from ab initio coupled-cluster theory," J. Phys. Chem. Lett. **13**, 7497–7503 (2022).
[19] H.-Z. Ye and T. C. Berkelbach, "Correlation-consistent gaussian basis sets for solids made simple," J. Chem. Theory Comput. **18**, 1595–1606 (2022).
[20] B. Hammer and J. Nørskov, "Electronic factors determining the reactivity of metal surfaces," Surf. Sci. **343**, 211–220 (1995).
[21] P. J. Feibelman, B. Hammer, J. K. Nørskov, F. Wagner, M. Scheffler, R. Stumpf, R. Watwe, and J. Dumesic, "The co/pt(111) puzzle," J. Phys. Chem. B **105**, 4018–4025 (2001).
[22] I. Grinberg, Y. Yourdshahyan, and A. M. Rappe, "Co on pt(111) puzzle: A possible solution," J. Chem. Phys. **117**, 2264–2270 (2002).
[23] A. Patra, H. Peng, J. Sun, and J. P. Perdew, "Rethinking co adsorption on transition-metal surfaces: Effect of density-driven self-interaction errors," Phys. Rev. B **100**, 035442 (2019).
[24] A. Stroppa and G. Kresse, "The shortcomings of semi-local and hybrid functionals: what we can learn from surface science studies," New J. Phys. **10**, 063020 (2008).
[25] T. Helgaker, W. Klopper, H. Koch, and J. Noga, "Basis-set convergence of correlated calculations on water," J. Chem. Phys. **106**, 9639–9646 (1997).
[26] K. A. Peterson, "Gaussian basis sets exhibiting systematic convergence to the complete basis set limit," in *Ann. Rep. Comput. Chem.*, Vol. 3 (Elsevier, 2007) pp. 195–206.
[27] J. G. Hill, "Gaussian basis sets for molecular applications," Int. J. Quantum Chem. **113**, 21–34 (2013).
[28] S. Lehtola, "Curing basis set overcompleteness with pivoted cholesky decompositions," J. Chem. Phys. **151**, 241102 (2019).
[29] J. Lee, X. Feng, L. A. Cunha, J. F. Gonthier, E. Epifanovsky, and M. Head-Gordon, "Approaching the basis set limit in gaussian-orbital-based periodic calculations with transferability: Performance of pure density functionals for simple semiconductors," J. Chem. Phys. **155**, 164102 (2021).
[30] L. E. Daga, B. Civalleri, and L. Maschio, "Gaussian basis sets for crystalline solids: All-purpose basis set libraries vs system-specific optimizations," J. Chem. Theory Comput. **16**, 2192–2201 (2020).
[31] M. Neef and K. Doll, "Co adsorption on the cu(111) surface: A density functional study," Surf. Sci. **600**, 1085–1092 (2006).
[32] J. Paier, C. V. Diaconu, G. E. Scuseria, M. Guidon, J. VandeVondele, and J. Hutter, "Accurate hartree-fock energy of extended systems using large gaussian basis sets," Phys. Rev. B **80**, 174114 (2009).
[33] J. VandeVondele and J. Hutter, "Gaussian basis sets for accurate calculations on molecular systems in gas and condensed phases," J. Chem. Phys. **127**, 114105 (2007).
[34] Y. Zhou, E. Gull, and D. Zgid, "Material-specific optimization of gaussian basis sets against plane wave data," J. Chem. Theory Comput. **17**, 5611–5622 (2021).
[35] W.-L. Li, K. Chen, E. Rossomme, M. Head-Gordon, and T. Head-Gordon, "Optimized pseudopotentials and basis sets for semiempirical density functional theory for electrocatalysis applications," J. Phys. Chem. Lett. **12**, 10304–10309 (2021).
[36] A. Erba, J. K. Desmarais, S. Casassa, B. Civalleri, L. Donà, I. J. Bush, B. Searle, L. Maschio, L. Edith-Daga, A. Cossard, C. Ribaldone, E. Ascrizzi, N. L. Marana, J.-P. Flament, and B. Kirtman, "Crystal23: A program for computational solid state physics and chemistry," J. Chem. Theory Comput. **19**, 6891–6932 (2023).
[37] T. H. Dunning, "Gaussian basis sets for use in correlated molecular calculations. i. the atoms boron through neon and hydrogen," J. Chem. Phys. **90**, 1007–1023 (1989).
[38] D. Bohm and D. Pines, "A collective description of electron interactions: Iii. coulomb interactions in a degenerate electron gas," Phys. Rev. **92**, 609–625 (1953).
[39] M. Gell-Mann and K. A. Brueckner, "Correlation energy of an electron gas at high density," Phys. Rev. **106**, 364–368 (1957).
[40] F. Furche, "Molecular tests of the random phase approximation to the exchange-correlation energy functional," Phys. Rev. B **64**, 195120 (2001).
[41] X. Ren, P. Rinke, C. Joas, and M. Scheffler, "Random-phase approximation and its applications in computational chemistry and materials science," J. Mater. Sci. **47**, 7447–7471 (2012).
[42] F. Weigend, F. Furche, and R. Ahlrichs, "Gaussian basis sets of quadruple zeta valence quality for atoms h-kr," J. Chem. Phys. **119**, 12753–12762 (2003).
[43] F. Weigend and R. Ahlrichs, "Balanced basis sets of split valence, triple zeta valence and quadruple zeta valence quality for h to rn: Design and assessment of accuracy," Phys. Chem. Chem. Phys. **7**, 3297–3305 (2005).
[44] F. Jensen, "Polarization consistent basis sets: Principles," J. Chem. Phys. **115**, 9113–9125 (2001).
[45] R. A. Kendall, T. H. Dunning, and R. J. Harrison, "Electron affinities of the first-row atoms revisited. systematic basis sets and wave functions," J. Chem. Phys. **96**, 6796–6806 (1992).
[46] D. E. Woon and T. H. Dunning, "Gaussian basis sets for use in correlated molecular calculations. v. core-valence basis sets for boron through neon," J. Chem. Phys. **103**, 4572–4585 (1995).
[47] W. A. de Jong, R. J. Harrison, and D. A. Dixon, "Parallel douglas-kroll energy and gradients in nwchem: Estimating scalar relativistic effects using douglas-kroll contracted basis sets," J. Chem. Phys. **114**, 48 (2001).
[48] P. Pollak and F. Weigend, "Segmented contracted error-consistent basis sets of double- and triple-ζ valence quality for one- and two-component relativistic all-electron calculations," J. Chem. Theory Comput. **13**, 3696–3705 (2017).
[49] M. F. Peintinger, D. V. Oliveira, and T. Bredow, "Consistent gaussian basis sets of triple-zeta valence with polarization quality for solid-state calculations," J. Comput. Chem. **34**, 451–459 (2013).
[50] N. B. Balabanov and K. A. Peterson, "Systematically convergent basis sets for transition metals. i. all-electron correlation consistent basis sets for the 3d elements sc-zn," J. Chem. Phys. **123**, 064107 (2005).
[51] A. Annaberdiyev, G. Wang, C. A. Melton, M. C. Bennett, L. Shulenburger, and L. Mitas, "A new generation of effective core potentials from correlated calculations: 3d transition metal series," J. Chem. Phys. **149**, 134108 (2018).
[52] M. C. Bennett, C. A. Melton, A. Annaberdiyev, G. Wang, L. Shulenburger, and L. Mitas, "A new generation of effective core potentials for correlated calculations," J. Chem. Phys. **147**, 224106 (2017).
[53] C. Cao, H. Q. Pham, Z. Guo, Y. Zhang, Z. Huang, X. Wen, J. Chen, and D. Lv, "Quantum many-body simulations of catalytic metal surfaces," (2025), arXiv:2508.13036 [physics.chem-ph].
[54] J. VandeVondele, M. Krack, F. Mohamed, M. Parrinello, T. Chassaing, and J. Hutter, "Quickstep: Fast and accurate density functional calculations using a mixed gaussian and plane waves approach," Comput. Phys. Commun. **167**, 103–128 (2005).
[55] H.-Z. Ye and T. C. Berkelbach, "*Ab Initio* surface chemistry with chemical accuracy: Application to water on metal oxides," arXiv preprint arXiv:2309.14640 (2023).
[56] K. E. Smyser, A. White, and S. Sharma, "Use of multigrids to reduce the cost of performing interpolative separable density fitting," J. Phys. Chem. A **128**, 7451–7461 (2024).
[57] B. T. G. Lau, B. Busemeyer, and T. C. Berkelbach, "Optical properties of defects in solids via quantum embedding with good active space orbitals," J. Phys. Chem. C **128**, 2959–2966 (2024).
[58] S. Banerjee and A. Y. Sokolov, "Non-dyson algebraic diagrammatic construction theory for charged excitations in solids," J. Chem. Theory Comput. **18**, 5337–5348 (2022).
[59] K. Laughon, J. M. Yu, and T. Zhu, "Periodic coupled-cluster green's function for photoemission spectra of realistic solids," J. Phys. Chem. Lett. **13**, 9122–9128 (2022).
[60] J. Li and T. Zhu, "Interacting-bath dynamical embedding for capturing nonlocal electron correlation in solids," Phys. Rev. Lett. **133**, 216402 (2024).
[61] P.-O. Widmark, P.-Å. Malmqvist, and B. O. Roos, "Density matrix averaged atomic natural orbital (ano) basis sets for correlated molecular wave functions," Theor. Chim. Acta **77**, 291–306 (1990).
[62] B. Kincaid, G. Wang, H. Zhou, and L. Mitas, "Correlation consistent effective core potentials for late 3d transition metals adapted for plane wave calculations," J. Chem. Phys. **157**, 174307 (2022).
[63] C. Hartwigsen, S. Goedecker, and J. Hutter, "Relativistic separable dual-space gaussian pseudopotentials from h to rn," Phys. Rev. B **58**, 3641–3662 (1998).

[64]W. Kutzelnigg and W. Liu, "Quasirelativistic theory equivalent to fully relativistic theory," J. Chem. Phys. **123**, 241102 (2005).
[65]W. Liu and D. Peng, "Exact two-component hamiltonians revisited," J. Chem. Phys. **131**, 031104 (2009).
[66]L. Cheng and J. Gauss, "Analytic energy gradients for the spin-free exact two-component theory using an exact block diagonalization for the one-electron dirac hamiltonian," J. Chem. Phys. **135**, 084114 (2011).
[67]J. P. Perdew, K. Burke, and M. Ernzerhof, "Generalized gradient approximation made simple," Phys. Rev. Lett. **77**, 3865–3868 (1996).
[68]J. Harl, L. Schimka, and G. Kresse, "Assessing the quality of the random phase approximation for lattice constants and atomization energies of solids," Phys. Rev. B **81**, 115126 (2010).
[69]L. Schimka, R. Gaudoin, J. c. v. Klimeš, M. Marsman, and G. Kresse, "Lattice constants and cohesive energies of alkali, alkaline-earth, and transition metals: Random phase approximation and density functional theory results," Phys. Rev. B **87**, 214102 (2013).
[70]N. K. Nepal, S. Adhikari, J. E. Bates, and A. Ruzsinszky, "Treating different bonding situations: Revisiting au-cu alloys using the random phase approximation," Phys. Rev. B **100**, 045135 (2019).
[71]L. Schimka, J. Harl, and G. Kresse, "Improved hybrid functional for solids: The hsesol functional," J. Chem. Phys. **134**, 024116 (2011).
[72]C.-N. Yeh, A. Shee, Q. Sun, E. Gull, and D. Zgid, "Relativistic self-consistent *gw*: Exact two-component formalism with one-electron approximation for solids," Phys. Rev. B **106**, 085121 (2022).
[73]N. Mardirossian and M. Head-Gordon, "ωb97x-v: A 10-parameter, range-separated hybrid, generalized gradient approximation density functional with nonlocal correlation, designed by a survival-of-the-fittest strategy," Phys. Chem. Chem. Phys. **16**, 9904–9924 (2014).
[74]P. E. Blöchl, "Projector augmented-wave method," Phys. Rev. B **50**, 17953–17979 (1994).
[75]G. Kresse and D. Joubert, "From ultrasoft pseudopotentials to the projector augmented-wave method," Phys. Rev. B **59**, 1758–1775 (1999).
[76]N.-K. Yu, J. Voss, and A. J. Medford, "Optimization of random phase approximation calculations for improved energies of molecules, solids, and surfaces," (2026), arXiv:2602.05138 [physics.chem-ph].
[77]B. X. Shi and T. C. Berkelbach, "Practical and accurate density functionals for transition-metal heterogeneous catalysis," (2026), arXiv:2602.14962 [cond-mat.mtrl-sci].
[78]X. Ren, P. Rinke, and M. Scheffler, "Exploring the random phase approximation: Application to co adsorbed on cu(111)," Phys. Rev. B **80**, 045402 (2009).
[79]L. Schimka, J. Harl, A. Stroppa, A. Grüneis, M. Marsman, F. Mittendorfer, and G. Kresse, "Accurate surface and adsorption energies from many-body perturbation theory," Nat. Mater. **9**, 741–744 (2010).
[80]R. Raval, S. Parker, M. Pemble, P. Hollins, J. Pritchard, and M. Chesters, "Ft-rairs, eels and leed studies of the adsorption of carbon monoxide on cu(111)," Surf. Sci. **203**, 353–377 (1988).
[81]E. J. Moler, S. A. Kellar, W. R. A. Huff, Z. Hussain, Y. Chen, and D. A. Shirley, "Spatial structure determination of $(\sqrt{3} \times \sqrt{3})r30^\circ$ and $(1.5 \times 1.5)r18^\circ$ co on cu(111) using angle-resolved photoemission extended fine structure," Phys. Rev. B **54**, 10862–10868 (1996).
[82]C.-R. Hsing, C.-M. Chang, C. Cheng, and C.-M. Wei, "Quantum monte carlo studies of co adsorption on transition metal surfaces," J. Phys. Chem. C **123**, 15659–15664 (2019).
[83]J. P. Carbone, A. Irmler, A. Gallo, T. Schäfer, W. Z. Van Benschoten, J. J. Shepherd, and A. Grüneis, "Co adsorption on pt(111) studied by periodic coupled cluster theory," Faraday Discuss. **254**, 586–597 (2024).
[84]S. Sharifzadeh, P. Huang, and E. Carter, "Embedded configuration interaction description of co on cu(111): Resolution of the site preference conundrum," J. Phys. Chem. C **112**, 4649–4657 (2008).
[85]S. F. Boys and F. Bernardi, "The calculation of small molecular interactions by the differences of separate total energies. some procedures with reduced errors," Mol. Phys. **19**, 553–566 (1970).
[86]B. X. Shi, A. Zen, V. Kapil, P. R. Nagy, A. Grüneis, and A. Michaelides, "Many-body methods for surface chemistry come of age: Achieving consensus with experiments," J. Am. Chem. Soc. **145**, 25372–25381 (2023).
[87]R. Shi, M.-Y. Zhang, P. Lin, L. He, and X. Ren, "Librpa: A software package for low-scaling first-principles calculations of random phase approximation electron correlation energy based on numerical atomic orbitals," Comput. Phys. Commun. **309**, 109496 (2025).